

Menu-Driven C Program for Multiplication Tables, Multiplication Grid and Consecutive Integers

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1. Problem Statement

Write a menu-driven C program that accepts two positive integers m and n , where $m \leq n$. Display the following menu and use a switch statement to perform the operation selected by the user:

1. Print the multiplication tables of m and n side by side, using multipliers from 1 to 10.
2. Display a multiplication grid containing the products of integers 1 to m and integers 1 to n .
3. For each integer from m to n , print that integer followed by the next five consecutive integers.
4. Exit the program.

Continue displaying the menu until the user selects the Exit option. Display an appropriate error message if the user enters an invalid menu choice.

2. Algorithm

1. Start.
2. Declare integer variables m , n and choice.
3. Read m and n from the user. Repeat the input step while $m \leq 0$, or $n \leq 0$, or $m > n$, displaying an appropriate error message each time.
4. Repeat steps 5 to 9 until the user selects choice = 4 (Exit).
5. Display the menu with four options: (1) Multiplication tables, (2) Multiplication grid, (3) Consecutive integers, (4) Exit.
6. Read the user's choice.
7. Evaluate choice using a switch statement:
 - case 1: For $i = 1$ to 10, print " $m \times i = m*i$ " and " $n \times i = n*i$ " side by side.
 - case 2: For $i = 1$ to m and $j = 1$ to n , compute product = $i*j$ and print it in grid form with row and column headers.
 - case 3: For $i = m$ to n , print i followed by $i+1$, $i+2$, $i+3$, $i+4$ and $i+5$.
 - case 4: Print an exit/goodbye message.
 - default: Print an "invalid choice" error message.
8. If choice $\neq 4$, return to step 5; otherwise exit the loop.
9. Stop.

3. Test-Case Table

S.No	Input (m , n / choice)	Menu Option	Expected Behaviour	Actual Result
1	$m=3$, $n=5$	1	Prints tables of 3 and 5 side by side, $i=1$ to 10 (e.g. $3 \times 1 = 3$, $5 \times 1 = 5$... $3 \times 10 = 30$, $5 \times 10 = 50$).	Pass — matches expected output
2	$m=3$, $n=5$	2	Prints 3x5 grid with column headers 1-5 and row headers 1-3; $\text{cell}(i,j) = i*j$.	Pass — matches expected output
3	$m=3$, $n=5$	3	Prints "3: 3 4 5 6 7 8", "4: 4 5 6 7 8 9", "5: 5 6 7 8 9 10".	Pass — matches expected output
4	$m=2$, $n=4$, choice=7	Invalid	Program should print an "Invalid choice!" error and redisplay the menu.	Pass — error message shown, menu re-displayed

S.No	Input (m, n / choice)	Menu Option	Expected Behaviour	Actual Result
5	m=6, n=2	—	Since $m > n$, the program should print an error and re-prompt for m and n.	Pass — re-prompted until m=4, n=8 entered
6	m=-3, n=5	—	Since $m \leq 0$, the program should print an error and re-prompt for m and n.	Pass — re-prompted until m=1, n=10 entered
7	choice = 4	4	Program prints a goodbye message and terminates the menu loop.	Pass — program exits cleanly

4. C Source Code

File name: menu_program.c

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/* =====
Program   : Menu-Driven Multiplication & Consecutive Integer Utility
Author    : Rudra Pratap Singh
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Description: Accepts two positive integers m and n ( $m \leq n$ ) from
              the user and repeatedly displays a menu that lets the
              user:
                1. Print multiplication tables of m and n side by side
                   (multipliers 1 to 10)
                2. Display a multiplication grid of (1..m) x (1..n)
                3. For each integer from m to n, print that integer
                   followed by the next five consecutive integers
                4. Exit the program
===== */

#include <stdio.h>

/* Function 1: Print multiplication tables of m and n side by side */
void printTables(int m, int n)
{
    int i;
    printf("\nMultiplication Tables of %d and %d (m=%d, n=%d)\n", m, n, m, n);
    for (i = 1; i <= 10; i++)
    {
        printf("%d x %-2d = %-6d\t%d x %-2d = %d\n", m, i, m * i, n, i, n * i);
    }
}

/* Function 2: Display multiplication grid of integers 1..m by 1..n */
void printGrid(int m, int n)
{
    int i, j;
    printf("\nMultiplication Grid (m=%d, n=%d)\n", m, n);

    printf("    ");
    for (j = 1; j <= n; j++)
        printf("%4d", j);
    printf("\n");
}

```

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printf("    ");
for (j = 1; j <= n; j++)
    printf("----");
printf("\n");

for (i = 1; i <= m; i++)
{
    printf("%2d |", i);
    for (j = 1; j <= n; j++)
        printf("%4d", i * j);
    printf("\n");
}
}

/* Function 3: For each integer i from m to n, print i and the next
five consecutive integers (i, i+1, i+2, i+3, i+4, i+5) */
void printConsecutive(int m, int n)
{
    int i, k;
    printf("\nConsecutive Integers from %d to %d (m=%d, n=%d)\n", m, n, m, n);
    for (i = m; i <= n; i++)
    {
        printf("%d:", i);
        for (k = 0; k <= 5; k++)
            printf(" %d", i + k);
        printf("\n");
    }
}

/* Function to display the main menu */
void displayMenu(void)
{
    printf("\n1. Multiplication tables of m and n\n");
    printf("2. Multiplication grid\n");
    printf("3. Consecutive integers\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
}

int main(void)
{
    int m, n, choice;

    /* Input validation: keep asking until m and n are positive
    integers and m <= n */
    do
    {
        printf("Enter two positive integers m and n (m <= n): ");
        if (scanf("%d %d", &m, &n) != 2)
        {
            printf("Invalid input. Please enter integers only.\n");
            while (getchar() != '\n')
                ; /* clear input buffer */
            m = 0; n = 0;
            continue;
        }
        if (m <= 0 || n <= 0)
            printf("Error: Both m and n must be positive integers.\n");
        else if (m > n)
            printf("Error: m must be less than or equal to n.\n");
    } while (m <= 0 || n <= 0 || m > n);

    do
    {

```

```

displayMenu();
if (scanf("%d", &choice) != 1)
{
    printf("Invalid input. Please enter a number between 1 and 4.\n");
    while (getchar() != '\n')
        ;
    choice = 0;
    continue;
}

switch (choice)
{
    case 1:
        printTables(m, n);
        break;
    case 2:
        printGrid(m, n);
        break;
    case 3:
        printConsecutive(m, n);
        break;
    case 4:
        printf("\nExiting the program. Goodbye!\n");
        break;
    default:
        printf("\nInvalid choice! Please enter a number between 1 and 4.\n");
}
} while (choice != 4);

return 0;
}

```

5. Compilation Command

The program was compiled using GCC on a Linux terminal with all warnings enabled:

```

$ gcc -Wall -o menu_program menu_program.c
$ ./menu_program

```

The program compiled with no errors or warnings.

6. Program Output (Execution Results)

6.1 Test 1 — $m = 3$, $n = 5$, options $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

Input sequence: 3 5 [Enter], 1 [Enter], 2 [Enter], 3 [Enter], 4 [Enter]

```

Enter two positive integers m and n (m <= n):
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice:
Multiplication Tables of 3 and 5 (m=3, n=5)
3 x 1 = 3      5 x 1 = 5
3 x 2 = 6      5 x 2 = 10
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50

```

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice:

Multiplication Grid (m=3, n=5)

	1	2	3	4	5
1	1	2	3	4	5
2	2	4	6	8	10
3	3	6	9	12	15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice:

Consecutive Integers from 3 to 5 (m=3, n=5)

3: 3 4 5 6 7 8

4: 4 5 6 7 8 9

5: 5 6 7 8 9 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice:

Exiting the program. Goodbye!

6.2 Test 2 — Invalid menu choice (m = 2, n = 4, choice = 7)

Input sequence: 2 4 [Enter], 7 [Enter], 4 [Enter]

Enter two positive integers m and n (m <= n):

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice:

Invalid choice! Please enter a number between 1 and 4.

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice:

Exiting the program. Goodbye!

6.3 Test 3 — Invalid input, m > n (m = 6, n = 2, then m = 4, n = 8)

Input sequence: 6 2 [Enter], 4 8 [Enter], 4 [Enter]

Enter two positive integers m and n (m <= n): Error: m must be less than or equal to n.

Enter two positive integers m and n (m <= n):

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice:

Exiting the program. Goodbye!

6.4 Test 4 — Invalid input, negative m (m = -3, n = 5, then m = 1, n = 10)

Input sequence: -3 5 [Enter], 1 10 [Enter], 4 [Enter]

```
Enter two positive integers m and n (m <= n): Error: Both m and n must be positive integers.  
Enter two positive integers m and n (m <= n):  
1. Multiplication tables of m and n  
2. Multiplication grid  
3. Consecutive integers  
4. Exit  
Enter your choice:  
Exiting the program. Goodbye!
```

7. Conclusion

A menu-driven C program was successfully designed, coded and tested to display multiplication tables of two numbers side by side, generate a multiplication grid, and list consecutive integers, using a switch statement to route the user's menu choice. Input validation ensures m and n are positive integers with $m \leq n$, and invalid menu selections are handled gracefully with an error message. All test cases listed in Section 3 produced the expected output, confirming that the program meets the given requirements.